

IMXRT1050 EVKB Board Hardware User's Guide

Contents

1. Introduction

This document is a Hardware User's Guide for the IMXRT1050 Evaluation Kit (EVKB) based on the NXP Semiconductor i.MX RT1050 Processor. This board is fully supported by NXP Semiconductor. The manual includes system setup and debugging, and provides detailed information on the overall design and usage of the EVKB board from a hardware systems perspective.

1.1. Board overview

This EVK board is a platform designed to showcase the most commonly used features of the i.MX RT1050 Processor in a small, low cost package. The IMXRT1050 EVKB board is an entry level development board, which gives the developer the option of becoming familiar with the processor before investing a large amount of resources in more specific designs.

1.	Introduction.....	1
1.1.	Board overview.....	1
1.2.	IMXRT1050 EVKB Contents.....	3
1.3.	IMXRT1050 EVKB Board revision history.....	3
2.	Specifications.....	3
2.1.	i.MX RT1050 Processor	5
2.2.	Boot Mode Configurations.....	5
2.3.	Power Tree.....	6
2.4.	SDRAM memory.....	8
2.5.	SD Card Slot.....	8
2.6.	Hyper Flash.....	8
2.7.	QSPI Flash.....	8
2.8.	Ethernet Connector	9
2.9.	USB PHY Connector	9
2.10.	Audio input / output Connector.....	9
2.11.	OpenSDA circuit (DAP-Link)	9
2.12.	JTAG Connector	10
2.13.	Arduino Expansion Port.....	10
2.14.	Camera Module Connector	11
2.15.	User Interface Switch.....	11
2.16.	Sensor	12
2.17.	User Interface LED Indicator.....	12
2.18.	LCD Interface	13
3.	PCB Information	13
4.	EVKB Design Files.....	13
5.	Contents of the Evaluation Kit.....	14
6.	Revision history	14

Features of the IMXRT1050 EVKB board are shown in [Table 1](#)

Table 1. Board features

Processor	NXP Processor	MIMXRT1052DVL6B(A1 silicon)
DRAM Memory	SDRAM 256 Mb, 166MHz	MT48LC16M16A2B4-6AIT: G
DCDC	MPS	MP2144GJ
LDO	UNION	UM1550S-18 UM1750S-00
Mass Storage	TF Card Slot	
	64 Mbit Quad SPI Flash	
	512 Mbit Hyper Flash	
Display Interface	LCD Connector	
Ethernet	10/100 Mbit/s Ethernet Connector. PHY Chip: KSZ8081RNB	
USB	USB 2.0 OTG Connector	
	USB 2.0 Host Connector	
Audio Connector	3.5 mm Audio Stereo Headphone Jack	
	Board-Mounted Microphone	
	Left & Right Speaker Out Connectors	
	SPDIF Interface(unpopulated)	
Power Connector	5V DC-Jack	
Debug Connector	JTAG 20-pin Connector (SWD by default)	
	OpenSDA with DAP-Link	
Sensor	FXOS8700CQ: 6-Axis Ecompass (3-Axis Mag, 3-Axis Accel)	
Camera	CMOS Sensor Interface	
CAN	CAN Bus Connector	
User Interface Button	ON/OFF, POR Reset, Reset, USER Button	
Led Indicator	Power Status, Reset, OpenSDA, USER LED	
Expansion Port	Arduino Interface	
PCB	3.937-inch x 5.9055-inch (10cm x 15cm), 4-layer board	

1.2. IMXRT1050 EVKB Contents

The IMXRT1050 EVKB contains the following items:

- IMXRT1050 EVKB Board
- USB Cable (Micro B)

1.3. IMXRT1050 EVKB Board revision history

- EVKB: Mass Product.

NOTE

EVKB Boards are based on A1 silicon.

2. Specifications

This chapter provides detailed information about the electrical design and practical considerations of the EVKB Board, and is organized to discuss each block in the [Figure 1](#) of the EVKB board.

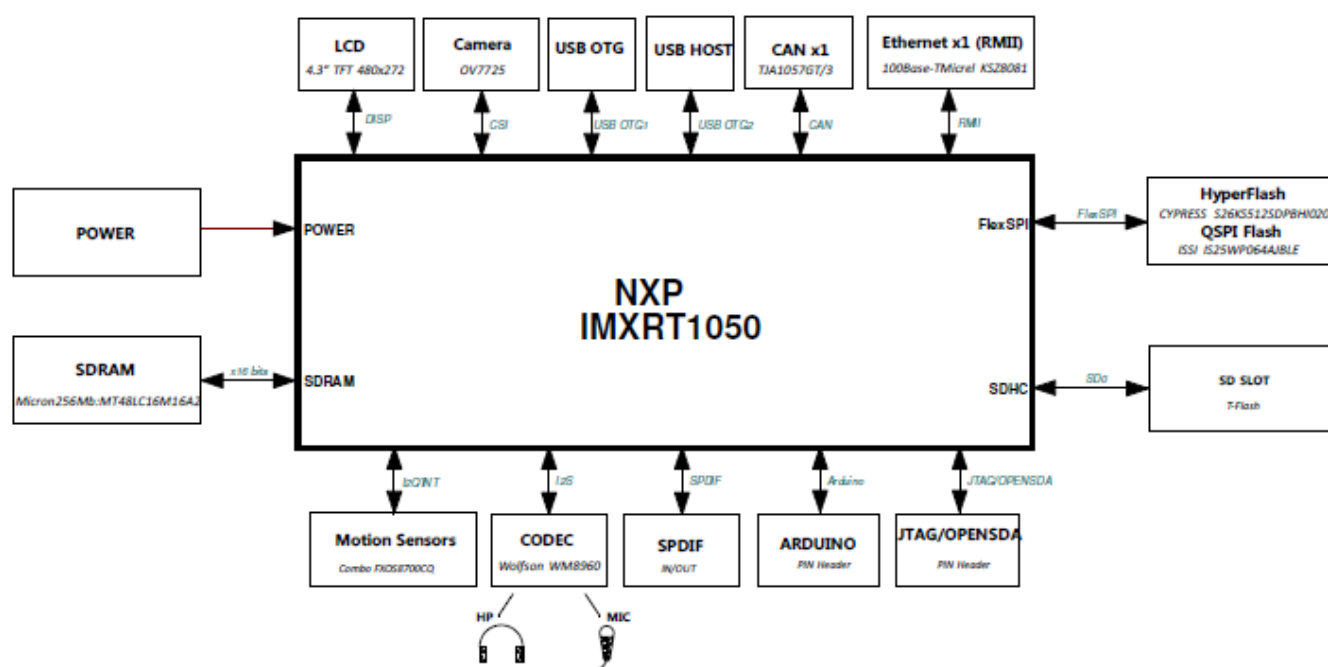


Figure 1. Block diagram

The overview of the IMXRT1050 EVKB Board is shown in [Figure 1](#) & [Figure 2](#).

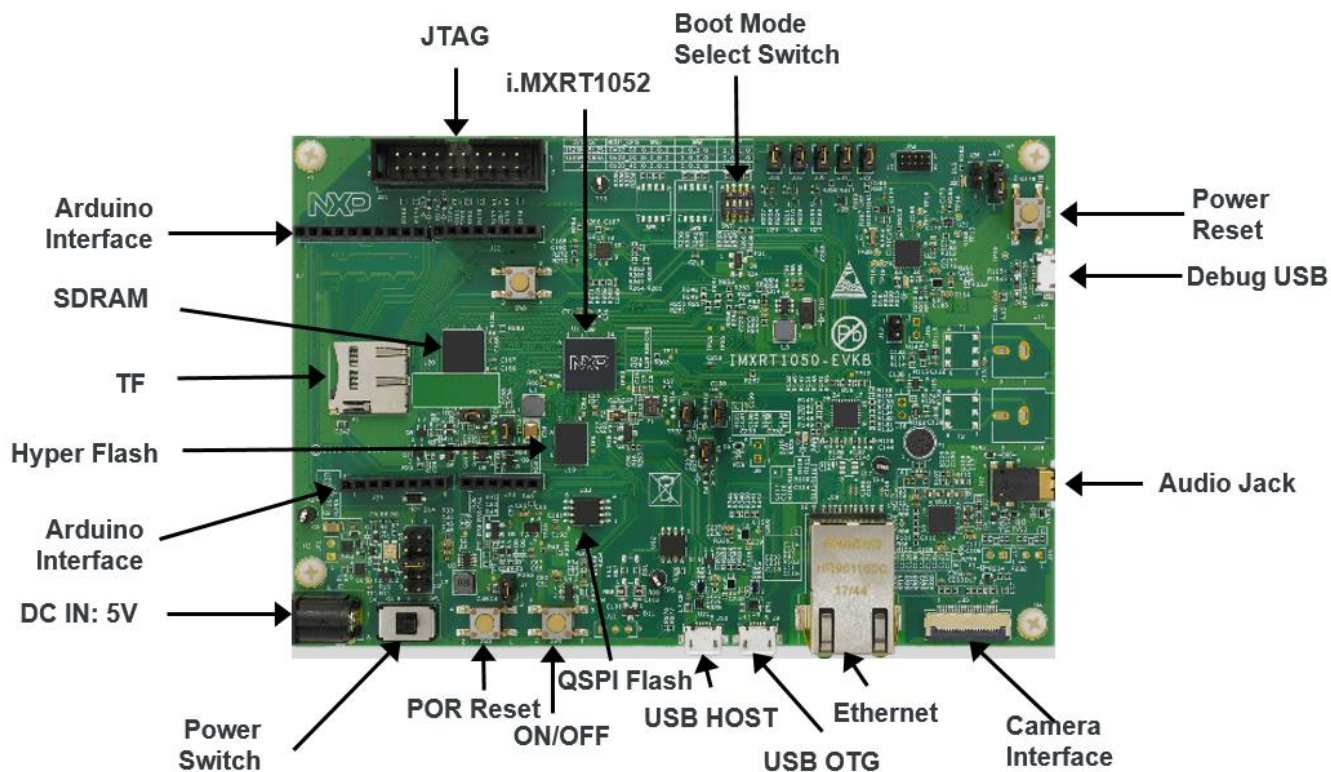


Figure 2. Overview of the IMXRT1050 EVKB Board (Front side)

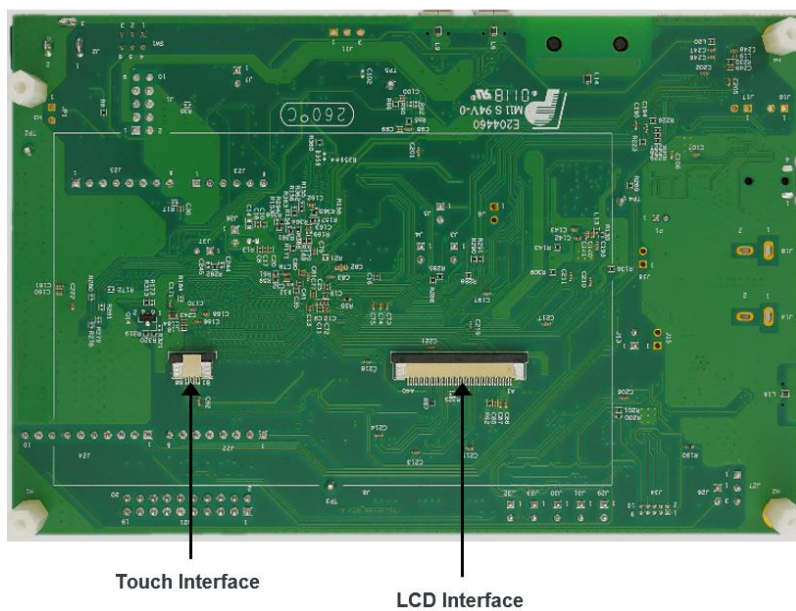


Figure 3. Overview of the IMXRT1050 EVKB Board (Back side)

2.1. i.MX RT1050 Processor

The i.MX RT1050 is a new processor family featuring NXP's advanced implementation of the ARM Cortex-M7 Core. It provides high CPU performance and best real-time response. The i.MX RT1050 provides various memory interfaces, including SDRAM, Raw NAND FLASH, NOR FLASH, SD/eMMC, Quad SPI, HyperBus and a wide range of other interfaces for connecting peripherals, such as WLAN, Bluetooth™, GPS, displays, and camera sensors. Same as other i.MX processors, i.MX RT1050 also has rich audio and video features, including LCD display, basic 2D graphics, camera interface, SPDIF and I2S audio interface.

The i.MX RT1050 applications processor can be used in areas such as industrial HMI, IoT, motor control and home appliances. The architecture's flexibility enables it to be used in a wide variety of other general embedded applications too. The i.MX processor provides all interfaces necessary to connect peripherals such as WLAN, Bluetooth™, GPS, camera sensors, and multiple displays.

The more detail information about i.MX RT1050 can be found in the [Datasheet and Reference manual](#).

2.2. Boot Mode Configurations

The device has four boot modes (one is reserved for NXP use). The boot mode is selected based on the binary value stored in the internal BOOT_MODE register. Switch (SW7-3 & SW7-4) is used to select the boot mode on the IMXRT1050 EVKB Board.

Table 2. Boot Mode pin settings

BOOT_MODE[1:0] (SW7-3 SW7-4)	BOOT Type
00	Boot From Fuses
01	Serial Downloader
10	Internal Boot
11	Reserved

Typically, the internal boot is selected for normal boot, which is configured by external BOOT_CFG GPIOs. The following [Table 3](#) shows the typical Boot Mode and Boot Device settings.

Table 3. Typical Boot Mode and Boot Device settings

SW7-1	SW7-2	SW7-3	SW7-4	Boot Device
OFF	ON	ON	OFF	Hyper Flash
OFF	OFF	ON	OFF	QSPI Flash
ON	OFF	ON	OFF	SD Card

NOTE

For more information about boot mode configuration, see the System Boot chapter of the [MIMXRT1050 Reference Manual](#).

For more information about IMXRT1050 EVKB boot device selection and configuration, see the [main board schematic](#).

2.3. Power Tree

A DC 5V external power supply is used to supply the IMXRT1050 EVKB Board at J2, and a slide switch SW1 is used to turn the Power ON/OFF. J28 and J9 also can be used to supply the EVKB Board. Different power supply need to configure different Jumper setting of J1. [Table 4](#) shows the details:

Table 4. Jumper settings of Power Supply

Power Supply	J1 Setting
J2	1-2
J9	3-4
J28	5-6

NOTE

For some computers' USB, it cannot support 500 ma before establishing communication. In this case, it is recommended to replace the computer or use the power adapter(J2) to power the EVKB Board.

The power tree is shown in the following figure.

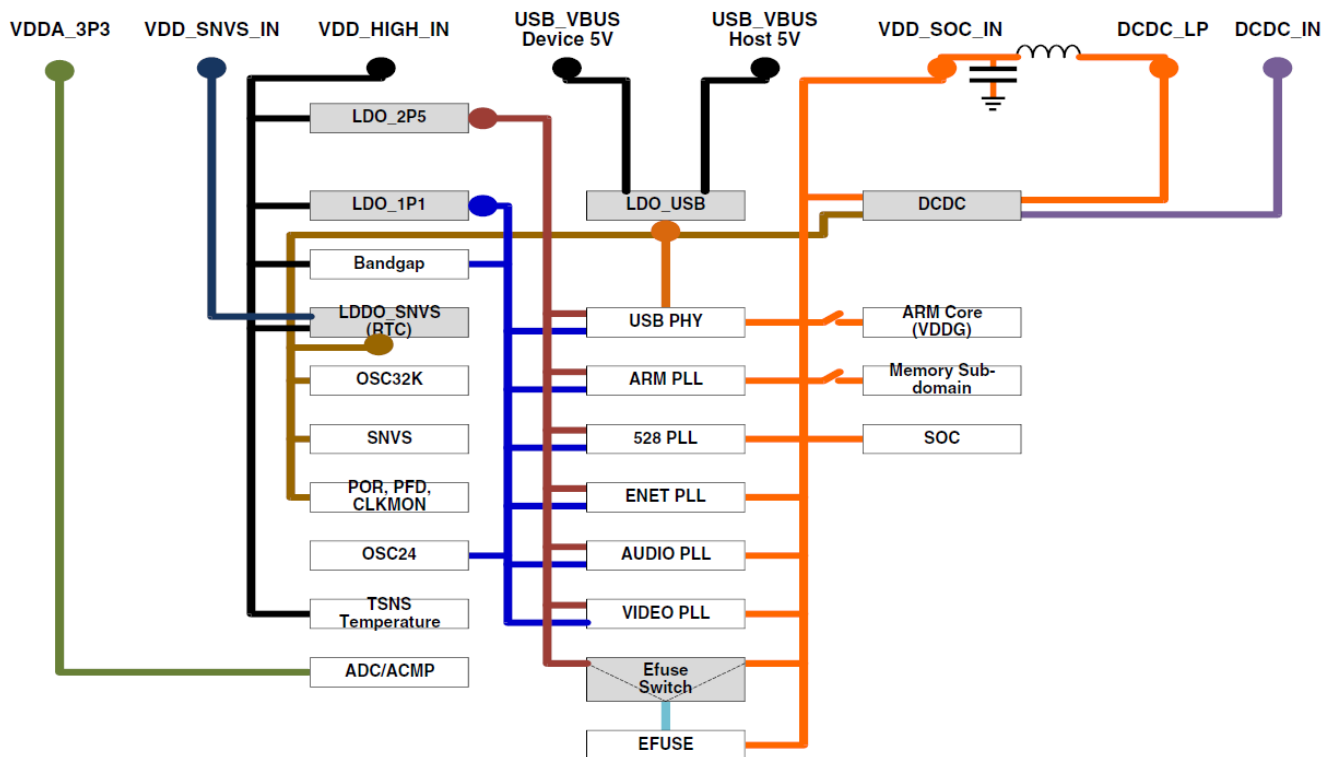


Figure 4. Power Tree

The power control logic of the IMXRT1050 EVKB board is shown in [Figure 5](#):

- It will power up SNVS firstly, then PMIC_REQ_ON will be switched on to enable external DC/DC to power up other power domains.
- ON/OFF button is used to switch ON/OFF PMIC_REQ_ON to control power modes.
- RESET button and WDOG output are used to reset the system power.

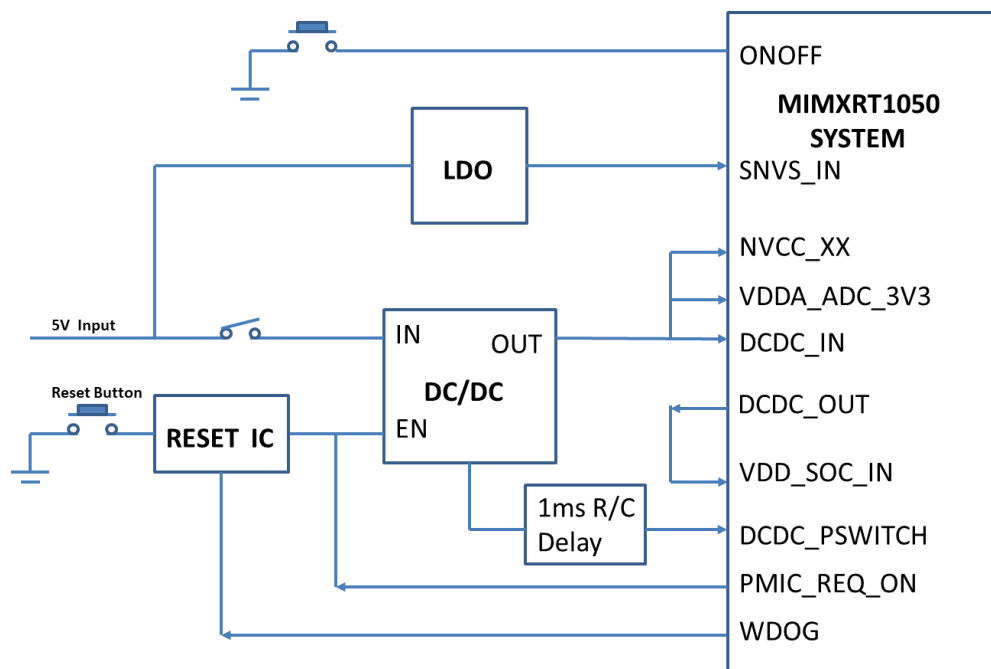


Figure 5. Power Control Diagram

The power rails on the board are shown in [Table 5](#).

Table 5. Power Rails

Power Rail	MIN (V)	TYP (V)	MAX (V)	Description
VDD_SOC_IN	0.925	--	1.26	Core supplies input voltage
VDD_HIGH_IN	3	3.3	3.6	VDD_HIGH_IN supply voltage
DCDC_IN	3 ¹	3.3 ¹	3.6 ¹	Power for DCDC
VDD_SNVS_IN	2.4	3	3.6	Power for SNVS and RTC
USB_OTG1_VBUS USB_OTG2_VBUS	4.4	5	5.5	Power for USB VBUS
VDDA_ADC		3.3	3.6	Power for 12-bit ADC

IMXRT1050 EVKB Board Hardware User's Guide, User's Guide, Rev. 0, 04/2018

	3			
NVCC_SD0	3	3.3	3.6	Power for GPIO in SDIO1 bank (3.3V mode)
	1.65	1.8	1.95	Power for GPIO in SDIO1 bank (1.8V mode)
NVCC_SD1	3	3.3	3.6	Power for GPIO in SDIO2 bank (3.3V mode)
	1.65	1.8	1.95	Power for GPIO in SDIO2 bank (1.8V mode)
NVCC_EMC	3	3.3	3.6	IO supply for GPIO in EMC bank (3.3 V mode)
	1.65	1.8	1.95	IO supply for GPIO in EMC bank (1.8 V mode)
NVCC_GPIO	3	3.3	3.6	IO power for GPIO

¹ For silicon A0, DCDC_IN voltage domain is 2.9V~3.1V

2.4. SDRAM memory

One 256 Mb, 166 MHz SDRAM (MT48LC16M16A2B4-6AIT: G) is used on the EVK Board.

2.5. SD Card Slot

There is a SD card slot(J20) on the IMXRT1050 EVKB Board.J20 is the Micro SD slot for USDHC1 interface. If the developer wants to boot from the SD Card, the boot device switch (SW7) settings should be: ON, OFF, ON, OFF, as shown in [Table 3](#).

2.6. Hyper Flash

On the IMXRT1050 EVKB Board, there is one 512 Mbit Hyper Flash device. If the developer wants to boot from the Hyper Flash, the boot device switch (SW7) settings should be: OFF, ON, ON, OFF, as shown in [Table 3](#).

2.7. QSPI Flash

A 64Mbit QSPI Flash is used on the IMXRT1050 EVKB Board. If the developer wants to boot from the QSPI Flash, the boot device switch(SW7) settings should be: OFF, OFF, ON, OFF, as shown in [Table 3](#).

By default, this QSPI Flash is disabled on the EVKB. To enable the onboard QSPI Flash, the settings need to be changed.

Step1:

Removed resistors: R356, R361 - R366.

Step2:

Weld 0 Ω resistors: R153 - R158.

2.8. Ethernet Connector

There is one Ethernet Mac controller in the MIMXRT1050 processor. The Ethernet subsystem of the IMXRT1050 EVKB Board is provided by the KSZ8081RNB 10/100M Ethernet Transceiver (U16) and a RJ45 (J19) with integrated Magnetic.

2.9. USB PHY Connector

The MIMXRT1050 contains 2 integrated USB 2.0 PHYs capable of connecting to USB host/device systems at the USB low-speed (LS) rate of 1.5 Mbits/s, full-speed (FS) rate of 12 Mbits/s or at the USB 2.0 high-speed (HS) rate of 480 Mbits/s.

2.10. Audio input / output Connector

The Audio CODEC used on the IMXRT1050 EVKB Board is Wolfson's Low Power, high quality Stereo Codec, WM8960. The IMXRT1050 EVKB Board include one headphone interface (J12), one onboard MIC (P1), two speaker interfaces (J16, J17), and the SPDIF interface (J14 & J18, DNP). J12 is a 3.5mm audio stereo headphone jack, which supports jack detect.

2.11. OpenSDA circuit (DAP-Link)

The OpenSDA circuit (CMSIS-DAP) is an open-standard serial and debug adapter. It bridges serial and debug communications between a USB host and an embedded target processor.

CMSIS-DAP features a mass storage device (MSD) bootloader, which provides a quick and easy mechanism for loading different CMSIS-DAP Applications such as flash programmers, run-control debug interfaces, serial-to-USB converters, and more. Two or more CMSIS-DAP applications can run simultaneously. For example, run-control debug application and serial-to-USB converter runs in parallel to provide a virtual COM communication interface while allowing code debugging via CMSIS-DAP with just single USB connection.

For the IMXRT1050 EVKB Board, J28 is the connector between the USB host and the target processor. Jumper to serial downloader mode to use stable DAP-Link debugger function. If developer wants to make OpenSDA going to the bootloader mode, J27 should jumper to 1-2, and press SW4 when power on. Meanwhile, the OpenSDA supports drag/drop feature for U-Disk. First, use the serial downloader mode and drag/drop the image file to U-Disk. Then select Hyper Flash as boot device and reset the Board, the image will run.

2.12. JTAG Connector

J21 is a standard 20-pin/2.54 mm Box Header Connector for JTAG. The pin definitions are shown in the [Figure 6](#). Support SWD by default.

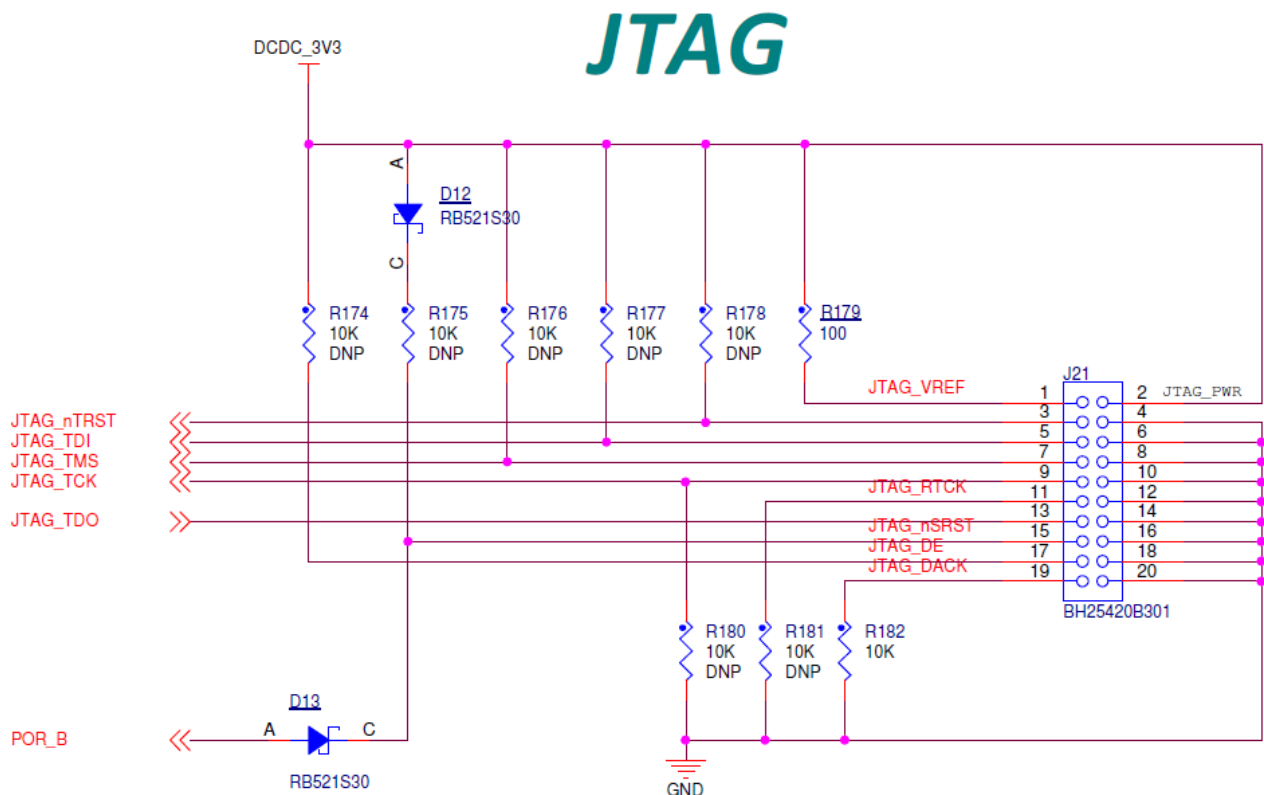


Figure 6. JTAG pin definitions

2.13. Arduino Expansion Port

J22 – J25 is defined as Arduino Interface. The pin definitions of Arduino Interface are shown in [Table 6](#).

Table 6. Arduino Interface pin definitions

J22	J23
UART_RX/D0	A0/ADC0
UART_TX/D1	A1/ADC1
D2/INT0	A2/ADC2
D3/INT1/PWM/OC2B	A3/ADC3
D4/T0/XCK	A4/ADC4/SDA
D5/TI/PWM	A5/ADC5/SCL

D6/AIN0/PWM/OC0A
D7/AIN1/PWM

J24	J25
D8/CLKO/ICP1	NC
D9/OC1A/PWM	IOREF
D10/SPI_CS	RESET
D11/OC2A/PWM/SPI_MOSI	3.3V
D12/SPI_MISO	5V
D13/SPI_CLK	GND
GND	GND
AREF	VIN
D14/I2C_SDA	
D15/I2C_SCL	

NOTE

J24 PIN9&PIN10 are not compatible with ARDUINO I2C, please use or wire to J23 PIN5&PIN6 instead and it will correct in next revision.

2.14. Camera Module Connector

One parallel CSI (Camera Sensor Interface) is supported by the i.MX RT1050. There is a Camera Module Connector (J35) on the IMXRT1050 EVKB Board. The CA031C based on OV7725 and CA111C based on MT9M114 can be used directly.

NOTE

J35 supports both MT9M114 and OV7725 camera module, but 3.3V is a violation to MT9M114 spec 3.1V. It proved fine for evaluation/demo with 3.3V supply, but in product design, it is recommended to adjust DCDC output or add level shifter.

2.15. User Interface Switch

There are four user interface switches on the IMXRT1050 EVKB Board. Their functionality is as below.

2.15.1. Power Switch

SW1 is a slide switch to control the power of the IMXRT1050 EVKB Board when the power supply is from J2. The function of this switch is listed below:

- Sliding the switch to the ON position connects the 5 V power supply to the Evaluation board main power system.
- Sliding the switch to OFF position immediately removes all power from the board.

2.15.2. ON/OFF Button

SW2 is the ON/OFF button for IMXRT1050 EVKB Board. A short pressing in OFF mode causes the internal power management state machine to change state to ON. In ON mode, a short pressing generates an interrupt (intended to be a software-controllable(power-down)). An approximate 5 seconds or more pressing causes a forced OFF. Both boot mode inputs can be disconnected.

2.15.3. Reset Button

There are two Reset Button on the EVK Board. SW3 is the Power On Reset Button. Pressing the SW3 in the Power On state will force to reset the system power except SNVS domain. The Processor will be immediately turn off and reinitiate a boot cycle from the Processor Power Off state. SW4 is Reset Button.

2.15.4. USER Button

SW8 is the USER Button(GPIO5-00) for developers using. Pressing can produce changes in high and low levels.

2.16. Sensor

U32 on the EVK Board is a 6-Axis Ecompass (3-Axis Mag, 3-Axis Accel) sensor FXOS8700CQ. The Ecompass is connected to i.MX RT1050 I2C1 port.

2.17. User Interface LED Indicator

There are four LED status indicators located on the EVK Board. The functions of these LEDs include:

- Main Power Supply(D3)
Green: DC 5V main supply is normal.
Red: J2 input voltage is over 5.6V.
Off: the board is not powered.
- Reset RED LED(D15)
- OpenSDA LED(D16)
- USER LED(D18)

2.18. LCD Interface

The enhanced Liquid Crystal Display Interface (eLCDIF) is a general purpose display controller.

The eLCDIF block supports the following:

- Displays that support moving pictures and require the RGB interface mode (DOTCLK interface).

The eLCDIF provides fully programmable functionality to supported interfaces:

- Bus master interface to source frame buffer data for display refresh.
- 8/16/18/24/32 bit LCD data bus support available depending on I/O mux options.
- Programmable timing and parameters for DOTCLK LCD interfaces.

If developers want to use LCD, NXP provides an optional LCD module RK043FN02H-CT which has a 4.3 inches touch-screen and supports a resolution of up to 480*3(GB)*272. This module contains two FPC cables. The LCD interface can be connected to J8(A1-A40) and the CPT interface can be connected to J8(B1-B6). LCD modules can be purchased from the [NXP website](#).

3. PCB Information

The IMXRT1050 EVKB Board is made using standard 4-layer technology. The material used was FR-4. The PCB stack-up information is shown in [Table 7](#).

Table 7. Board stack-up information

Layer	Description	Copper(Oz)	Dielectric Thickness(mil)
1	Signal	1	—
	Dielectric	—	3
2	GND	1	—
	Dielectric	—	52
3	Power	1	—
	Dielectric	—	3
4	Signal	1	—

4. EVKB Design Files

The schematics, layout files, and gerber files (including Silkscreen) can be downloaded from nxp.com/MIMXRT1050-EVK

5. Contents of the Evaluation Kit

Table 8. EVK contents

Item	Description
EVK Board	EVKB Board with processor, memory, interfaces, etc
USB Cable	USB cable (Micro-B to Standard-A)

NOTE

Power adaptor, Micro SD Card, LCD Module and Camera Module are not standard parts of the Evaluation Kit.

6. Revision history

Table 9 summarizes the changes made to this document since the initial release.

Table 9. Revision history

Revision number	Date	Substantive changes
0	04/2018	Initial release

How to Reach Us:

Home Page:
nxp.com

Web Support:
nxp.com/support

Information in this document is provided solely to enable system and software implementers to use NXP products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits based on the information in this document. NXP reserves the right to make changes without further notice to any products herein.

NXP makes no warranty, representation, or guarantee regarding the suitability of its products for any particular purpose, nor does NXP assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters that may be provided in NXP data sheets and/or specifications can and do vary in different applications, and actual performance may vary over time. All operating parameters, including "typicals," must be validated for each customer application by customer's technical experts. NXP does not convey any license under its patent rights nor the rights of others. NXP sells products pursuant to standard terms and conditions of sale, which can be found at the following address: nxp.com/SalesTermsandConditions.

NXP, the NXP logo, NXP SECURE CONNECTIONS FOR A SMARTER WORLD, Freescale, the Freescale logo are the trademarks of NXP B.V. All other product or service names are the property of their respective owners.

Arm, the Arm logo, and Cortex are registered trademarks of Arm Limited (or its subsidiaries) in the EU and/or elsewhere.. All rights reserved.

© 2018 NXP B.V.

Document Number: IMXRT1050EVKBHUG
Rev. 0
04/2018

